

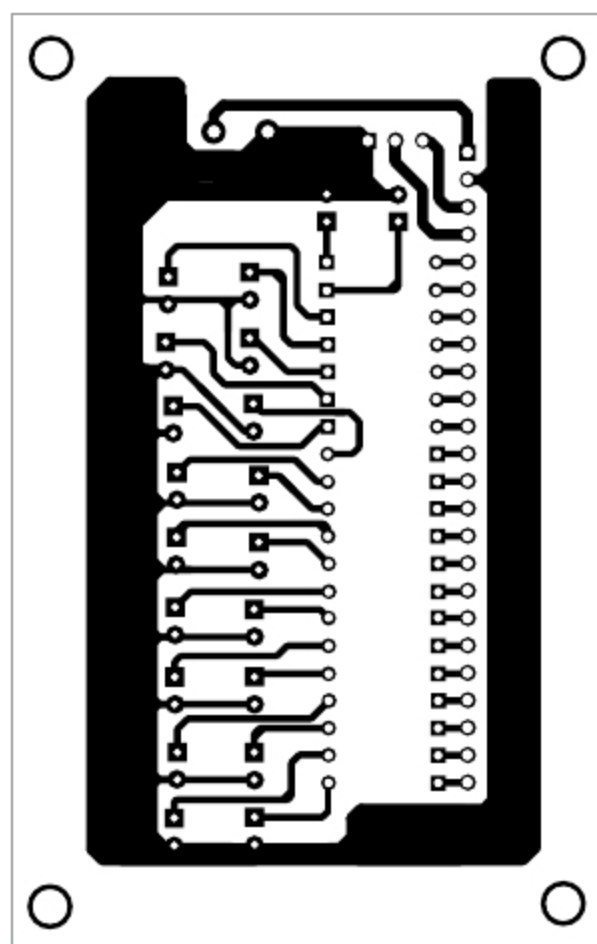


Fig. 2: Actual-size PCB layout of the LED flasher

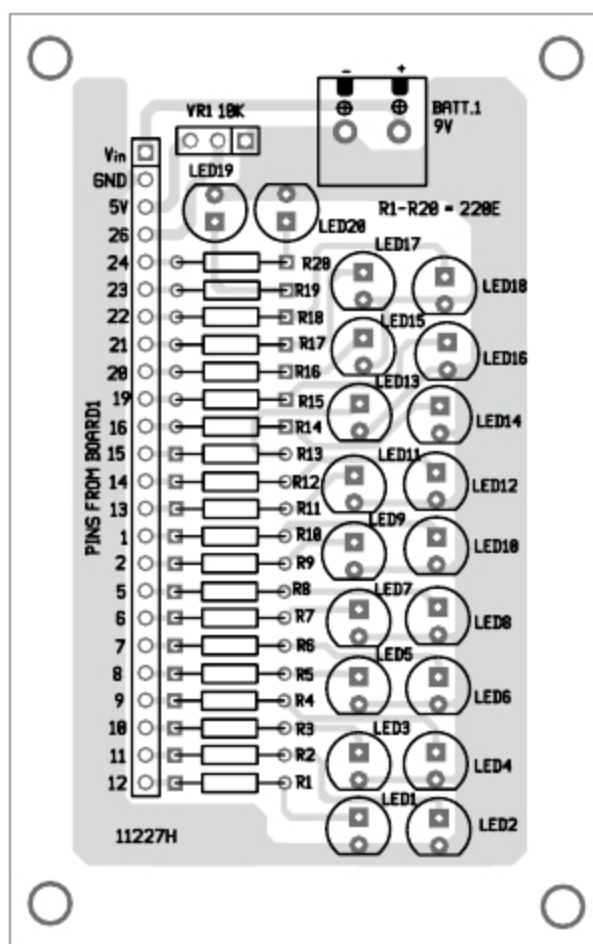


Fig. 3: Components layout for the PCB

Unlike Arduino Uno, Arduino Nano does not have the DC power jack for external power source.

The software for the flasher is written in Arduino programming language/sketch. Arduino Nano is programmed using Arduino IDE. Select the correct board from Board→Tools menu in Arduino IDE, select COM port and upload the program (Flasher.ino) through the USB port in the computer.

Construction and testing

Connect LEDs (LED1 through LED20) to the port pins of Arduino Nano, as per the circuit diagram. You will need a breadboard, veroboard or designed PCB for connecting the resistors and LEDs to Arduino Nano.

An actual-size PCB layout of the circuit is shown in Fig. 2 and its components layout in Fig. 3.

After assembling the circuit on the designed PCB and programming

PARTS LIST

Semiconductors:

Board1 - Arduino Nano
LED1-LED20 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1-R20 - 220-ohm
VR1 - 10-kilo-ohm potmeter

Miscellaneous:

BATT.1 - 9V battery
- 24-pin Berg strip female connector

(All these are available from kitsnspares.com)

Arduino Nano, connect 9V DC or 9V battery to the circuit. You will notice that all LEDs (LED1 through LED20) start to glow randomly, one-by-one. The LED flashing pattern repeats in this fashion till power supply is switched off.

Many LED strings can be used in place of a single LED using suitable drivers. However, total current of Arduino Nano is limited to 200mA, so it is advisable to use a suitable driver if more than one LED is to be lit at a time. Bi-colour or RGB LEDs can be used instead of single-colour LEDs. **EFY**



A. Samiuddin is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers and Arduino programming

Wanna Develop Software...
Faster, Better & Cheaper?
DO OPEN SOURCE.



16th Edition

OPEN
SOURCE INDIA

17th & 18th October, 2019
NIMHANS Bengaluru

Asia's #1 Conference
On Open Source

Mark the Dates on Your Calendar.

Registrations Open at
www.OpenSourceIndia.in

Want to Engage with India's Best Software Developers & IT Managers? Exhibit at OSI.
www.OpenSourceIndia.in
osi@efy.in | +91-98111 55335 |

EFY Note

The source code of this project is available for free download at source.efymag.com